Niccolò Covoni

Assistant Professor · Sophia University Institute

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Area of Research

Philosophy of Information, Philosophy of Physics, Philosophy of Science, and Metaphysics

Academic Positions

2027 - 2030 **Fixed-Term Assistant Professor**, Sophia University Institute — *Loppiano, Italy*

Education

University of Svizzera Italiana USI — *Lugano, Switzerland*

PhD in Philosophy — *2023 - 2026*

- Supervisor: Claudio Calosi

University of Urbino "Carlo Bo - Department of Pure and Applied Sciences — *Urbino, Italy*

PhD in Research Methods in Science and Technology — *2023 - 2026*

- Title of project: The Reality of Entropy: foundational study of entropy and information
- Supervisor: Vincenzo Fano

University of Firenze — *Florence, Italy*

MS in Logic, Philosophy and History of Science — *2020 - 2023*

- Title of dissertation: Methods of Abstraction in Information Theory The Case of Quantum Like Machine Learning
- Supervisor: Silvano Zipoli Caiani
- Co-Supervisor: Giuseppe Sergioli
- Graduation Mark: 110/110 cum Laude

University of Pisa *Pisa, Italy*

BS in Philosophy *2017 - 2020*

- Title of dissertation: Considerations on the notion of Birkhoff-Von Neumann's Quantum Logic
- Supervisor: Luca Bellotti
- Graduation Mark: 107/110

Publications

Books

Carlo Rovelli, Niccolò Covoni, Andrea Di Biagio. *Tractatus Quanticus*. Wiley. Forthcoming April 2027. 176 pp. ISBN 978-1-394-44530-1. Publisher page

Niccolò Covoni, Andrea Di Biagio, Carlo Rovelli. *Tractatus Quanticus*. Adelphi. Italian edition under contract.

Forthcoming

Niccolò Covoni, Marco Sanchioni. Forthcoming. Entanglement, Relative Facts, and the Meaning of Information in Relational Quantum Mechanics, in *European Journal for Philosophy of Science*. Accepted version

Published

Niccolò Covoni, Marco Sanchioni. 2026. Relationality without Agents: Quantum Correlations and the Limits of QBism, in Anguel S. Stefanov and Vera Lyubenova (eds.), *Niels Bohr and Contemporary Philosophy of Science*, Minkowski Institute Press, Montreal, pp. 61-86. ISBN 978-1-998902-81-1.

Vincenzo Fano, Niccolò Covoni. 2026. Phenomenological Elements in Carnap's *Aufbau*: The Epistemological Role of Sensible Intuition, in *Sophia*, XVIII (2026-1), 29-51.

Niccolò Covoni, Marco Sanchioni. 2026. Relational Information Towards a New Kind of Information in Quantum Mechanics in *Foundations of Physics*, 56(1): 14.

Niccolò Covoni, Vincenzo Fano, Marco Sanchioni. 2025. Relational Quantum Mechanics and (Soft) Perspectivism in *Foundations of Physics*, 56(1): 6.

Niccolò Covoni, Carlo Rovelli. 2025. Tractatus Quanticus, https://arxiv.org/abs/2512.06034

Niccolò Covoni, Gino Tarozzi. 2025. Tra realismo e neopositivismo:l'eredità di Ludovico Geymonat e Evandro Agazzi nella filosofia della scienza italiana in *Bíos theôrêtikós e bíos praktikós. Studi offerti a Fabio Minazzi per i suoi settant'anni*, Mimesis Edizioni, Milano.

Niccolò Covoni, Flavia Marcacci, Gino Tarozzi. 2025. From neo-idealistic historicism to Geymonat's materialistic and Agazzi's objectivistic realism: the roots of the Italian

philosophy of science, in *BIOETHICS UPDATE*, 11(2), https://doi.org/10.24875/BUP.25000013.

Niccolò Covoni. 2025. From Shannon to Von Neumann: A partial understanding of Information, in *Atti del XLIV Congresso Nazionale SISFA - Firenze 2024*, 43(10), 89-94.

Niccolò Covoni, Giovanni Galli. 2025. A (quasi)-Mathematical Theory of Public Language, in Proceedings 46th International Wittgenstein Symposium 2025.

Niccolò Covoni, Giovanni Macchia, Davide Pietrini and Gino Tarozzi. 2025. Non-standard realistic models of quantum phenomena and new forms of complementarity, in *Models and Representations in Science, edited by Hans-Peter Grosshans. Comptes Rendus de l'Académie Internationale de Philosophie des Sciences* 3, 39-54.

Niccolò Covoni, Alberto Corti, Vincenzo Fano. 2024. A Model-Based form of Naturalised Metaphysics, Revista Portuguesa de Filosofia, 80(4), 783-818.

Niccolò Covoni, Gino Tarozzi. 2024. Empiricism and Realism Reconciled in Agazzi's Conception of Scientific Objectivity, Distinctio: Journal of Intersubjective Studies, Zagreb, 3(1), 47-55.

Niccolò Covoni, Silvano Zipoli Caiani. 2024. Levels of Abstraction as family resemblances from the classical to the quantum mechanical representation of reality in Proceedings 45th International Wittgenstein Symposium 2024 "Facets of Reality — Contemporary Debates"

Marco Ciardi, Niccolò Covoni. 2024. La profezia di Ada Lovelace. L'intelligenza artificiale prima di Turing. On Filosofia dell'intelligenza artificiale. Sfide etiche e teoriche, Il Mulino, Bologna, ISBN: 978-88-15-38737-0

Marco Ciardi, Niccolò Covoni. 2023. Dalla mano per calcolare ai calcolatori per la mano. Sull'origine del pensiero numerico, Guide Museo e Istituto fiorentino di Preistoria, ISBN: 979-12-80445-06-3.

In Review

Niccolò Covoni, Hanna Podsędkowska, Rafał Wieczorek. 2026. Segal Entropy and Relational Information

Alberto Corti, Niccolò Covoni, Vincenzo Nespeca, Marco Sanchioni. 2026. Progressive Scale Realism A naturalized realistic interpretation of Spin

Niccolò Covoni, Marco Sanchioni. 2026. Progress Crucial Realism

Niccolò Covoni, Marco Sanchioni. 2026. Progress-Crucial Realism Toward a Selective Meta-Ontology

Niccolò Covoni, Marco Sanchioni. 2026. From “It from Bit” to Relational Quantum Mechanics: Wheeler, Rovelli, and the Relational Turn in Quantum Foundations

Prizes & Grants

01/2025 - 03/2025	**Scholarship from The QISS consortium**, University of Western Ontario	*€3,000*
from 1/11/2023	**PhD Scolarship founded by Department of Pure and Applied Sciences (DiSPeA)**, University of Urbino ‘Carlo Bo’	*€16,243,00 per year*
2020-22	**Scholarship at ”Percorso di Eccellenza Trasversale 2020/21” as Excellent Student**, University of Florence	*€4,000*

Presentations

Invited Talks

27–30/05/2027. **Invited speaker, Foundations of Quantum Theory session**. New Directions in the Foundations of Physics, Hotel Bohinj, Slovenia.

23/04/2026. **Relational Information Towards a New Kind of Information in Quantum Mechanics**.Weekly Meeting Departamento de Fisica Teorica, Universidad Complutense de Madrid, Madrid, Spain.

19/04/2026. **Entanglement, fatti relativi e il significato di ”informazione” nell’ interpretazione relazionale della meccanica quantistica**. Il Paradosso della realta’ fisica 1985 -225, University of Urbino, Cesena, Italy.

11/03/2026. **Relational Information Towards a New Kind of Information in Quantum Mechanics**. University of Jagellonian, Krakow, Poland.

05/03/2026. **Relational Information Towards a New Kind of Information in Quantum Mechanics**. Katedra Teorii Prawdopodobieństwa i Statystyki WMiI UŁ wraz z Oddziałem Łódzkim PTM, University of Lodz, Lodz, Poland.

17/12/2025. **Once Again on von Neumann Entropy: Is It Always the Case That von Neumann Entropy Corresponds to Thermodynamic Entropy?**. Philosophical Foundations of Thermodynamics, Pontificia Universidad Católica de Valparaíso, Valparaíso, Chile.

20/11/2025. **Relational Information: Towards a New Type of Information in Quantum Mechanics**. Physics Meets Philosophy in Vienna, Institute for Quantum Optics and Quantum Information, Vienna, Austria.

Contributed Presentations

Entanglement, Relative Facts, and the Meaning of Information in Relational Quantum Mechanics
23/09/2026. The Warsaw Dialogues in the Philosophy of Physics 2026, Warsaw University of Technology, Warsaw, Poland

Relational Information Towards a New Kind of Information in Quantum Mechanics
22/07/2026. Asian Philosophy of Science Association (APSA) Conference 2026, NTU Singapore, Singapore

Entanglement, Relative Facts, and the Meaning of Information in Relational Quantum Mechanics
29/06/2026. SILFS Triennial International Conference 2026: Thinking the Future with Logic and the Philosophy of Sciences, Scuola Universitaria Superiore IUSS, Pavia, Italy

Segal Entropy and the notion of Information
09/05/2026. Workshop: Perspectives on the Foundations of Information Theory and Statistical Mechanics Università della Svizzera Italiana, Lugano, Switzerland

From 'It from Bit' to Relational Quantum Mechanics: Wheeler and Rovelli in Historical and Conceptual Perspective
10/12/2025. Towards a Quantum Culture: The Different Interpretations of Quantum Mechanics, Workshop of the Italian Society for the History of Physics, Università Studi di Milano, Milano, Italy

Relational Quantum Mechanics and (Soft) Perspectivism
16/10/2025. Philosophy of Science Conference on the Occasion of the 140th Anniversary of Niels Bohr's Birth, Sofia, Bulgaria

A (quasi)-Mathematical Theory of Public Language
12/08/2025. 46th International Wittgenstein Symposium, Kirchberg am Wechsel, Austria

Progress Crucial Realism
11-13/06/2025. SILFS Postgraduate Conference, Turin, Italy

From Shannon to Von Neumann: A partial understanding of Information
18/09/2024. XLIV National Congress of SISFA, Firenze, Italy

The Forbidden Realm for the Maxwell's Demon: Information Entropy
09/09/2024. Synergia PhD Research Symposium - Charting New Frontiers in Research

Levels of Abstraction as family resemblances: from the classical to the quantum mechanical representation of reality
12/08/2024. 45th International Wittgenstein Symposium, Kirchberg am Wechsel, Austria

Exploring Quantum Information Processes through the Method of Abstraction: Quantum-Levels of Abstraction
04/08/2024. XXV World Congress of Philosophy, Rome, Italy

Fearing Bitcoin
03/08/2024. XXV World Congress of Philosophy, Society Section SILFS, Rome, Italy

Method of Abstraction: A Tentative Exploration of Quantum Information Processes
07/12/2023. Seminar organized by SYNERGIA the Research Group in Logic, Language, Cognition, History and Phylosophy of Science, Università di Urbino Carlo BO, Urbino, Italy.

Contextuality of Abstraction
31/03/2023. Seminar organized by the professor of Philosophy of Mind Silvano Zipoli Caiani. University of Firenze, Firenze, Italy.

Una breve panoramica della Filosofia dell'informazione: dall'informazione classica all'informazione quantistica
10/11/2022. Seminar organized by the student university association Gavagai. Pisa, Italy.

Poster Presentations

Progress Crucial Realism
29/08/2025. 10th biennial meeting of the European Philosophy of Science Association. Groningen, Netherlands.

Progress Crucial Realism
01/07/2025. Foundation2025. Gdańsk, Poland.

Shannon's approach to quantum information
14/10/2024. Probability in Computer Science (PICS) 2024. Copenhagen, Denmark.

Public Presentations

Certo per Incerto: Quanta Informazione
20/04/2026. Facoltà di Teologia di Lugano. Lugano, Switzerland.

Teaching Experience

2026/2027	**Lugano Philosophy Colloquia**, Course for master's students	*Università della Svizzera italiana (USI), Lugano*
2026/2027	**Philosophy of Language**, Course for bachelor's students	*Sophia University Institute*
2026/2027	**Logic, Game Theory and Happiness**, Course for master's students; co-taught with Marco Sanchioni	*Sophia University Institute*
2026	**Introduction to the Philosophy of Thermodynamics**, Introductory course for philosophy students	*Università della Svizzera italiana (USI), Lugano*
2025/2026	**Philosophy of Language**, Teaching Assistant in the course 'Philosophy of Language' taught by Professor Marco Sanchioni	*Sophia University Institute*
09/05/2024	**Introduction to Edutainment** , Lecture for master students of Communication Science taught by professor Silvano Zipoli Caiani	*University of Florence*
02/2024 - 05/2024	**Lectures about Epistemology**, Teaching Assistant in the course 'Epistemology' taught by professor Gino Tarozzi	*University of Urbino*
15/12/2023	**Introduction to Description Logic**, Lecture for master students in Philosophy, in the course 'Web's Semantics' taught by professor Pierluigi Graziani	*University of Urbino*

Research Experience

07/26 - 08/26	**FTG Visiting Fellow**, Hosted by Professor Eugene Y. S. Chua, Foundations of Thermodynamics Group, Nanyang Technological University	*Singapore*

02/26 - 04/26	**Visiting Student**, Supervised by Professor Hanna Podsędkowska, Faculty of Mathematics and Computer Science, University of Lodz	*Lodz, Poland*
01/25 - 03/25	**Visiting Student**, Supervised by Professor Francesca Vidotto, Rotman Institute of Philosophy, Western University	*London, ON, Canada*
2024	**Visiting Student**, Supervised by Professors Roberto Giuntini and Giuseppe Sergioli, University of Cagliari	*Cagliari, CA, Italy*

Attendeed Events

Conferences and Workshops

Workshop: Exploring Quantum Technology and the Philosophy of Quantum Mechanics, 22/08/2025, TU Delft, Delft, Netherlands

QISS 2025 Conference, 14/04/2025-17/04/2025, Institute for Quantum Optics and Quantum Information (IQOQI), Wien, Austria

Workshop: Theoretical Equivalence and Related Topics, 08/04/2025, Università Svizzera Italiana, Lugano, Switzerland

4th Ca'Foscari-Lugano Workshop in Analytic Metaphysics, 16/12/2024, University Ca' Foscari, Venice, Italy

Workshop on Scientific Realism, 08/11/2024, University of Roma Tre, Rome, Italy

Ninth Annual Conference of the International Association for the Philosophy of Time, 24/06/2024-28/06/2024, University Svizzera Italiana, Lugano, Switzerland

Philosophy of Spontaneous Collapse, 20/05/2024-21/05/2024, University Svizzera Italiana, Lugano, Switzerland

Buchi, ombre, specchi, errori. Trent'anni di metafore nella filosofia di Achille Varzi, 02/05/2024, University Ca'Foscari, Venice, Italy

Variations in Philosophy of Mathematics, 13/04/2024-14/04/2024, University Svizzera Italiana, Lugano, Switzerland

Unification of Types and Multi-Level Modeling, 13/03/2024, King's College, London, United Kingdom

Formal Modeling of Ignorance, 07/09/2023 – 08/09/2023, Università degli Studi di Urbino Carlo BO, Urbino Italy

Triennial International Conference Logic, Philosphy of Science, and Global Challenges, 04/09/2023 – 07/09/2023, SILFS, Urbino, Italy

Linguaggio, Logica e Ontologia, 14/06/2023 – 15/06/2023, University of Pisa, Pisa, Italy

Information and Computing in Nature Philosophical Perspectives, 01/12/2022, Polish Academy of Arts and Sciences Commission for the Philosophy of Science, Online

Online Course 8 consciousness, 12/08/2022 – 19/08/2022, Steven Gouveia, Online

The 22nd Trends in Logic workshop, entitled Strong and Weak Kleene Logics, 18/07/2022 – 20/07/2022, ALOPHIS research group of the Department of Pedagogy, Psychological Sciences and Philosophy (Univestiy of Cagliari), Online

The manifest image and the scientific image: objects, properties and relations,29/11/2021 – 30/11/2021, University of Firenze, Florence, Italy

The Nature of Quantum Objects,11/11/2021 – 13/11/2021, University of Geneva, Geneva, Switzerland

PM65 An international workshop of logic to celebrate Pierluigi Minari's 65th birthday,15/10/2021, University of Firenze, Florence, Italy

Winter Schools

Probability in Computer Science (PICS) 2024 5th Ph.D. school in the series of Foundations of Programming and Software Systems, 14/10/2024 - 18/10/2024, It-Universitet, Copenhagen,Denmark, 2 ECTS.

Summer Schools

14th Ludwig Wittgenstein Summer School, 01/08/2023 – 05/08/2023, Austrian Ludwig Wittgenstein Society (ALWS), Kirchberg am Wechsel, Austria

XXV International Summer School in Philosophy of Physics, 06/06/2022 – 11/06/2022, Università degli Studi di Urbino Carlo BO, Urbino, Italy

12th Ludwig Wittgenstein Summer School, 03/08/2021 – 07/08/2021, Austrian Ludwig Wittgenstein Society (ALWS), Klagenfurt am Wörthersee, Austria

XXV International Summer School in Philosophy of Physics, 07/06/2021 – 11/06/2021, Università degli Studi di Urbino Carlo BO, Urbino, Italy

Service to the Profession

Research Coordination

Coordinator, CI^2 – Complexity & Information Initiative, International Center for Transdisciplinary Studies (ICTS), Sophia University Institute; with Marco Sanchioni as co-coordinator. 2026 - present.

Active Organization of Events

XXXI International Summer School in Philosophy of Physics - History and Philosophy of the Wave Function
15/06/2026 - 19/06/2026
Organiser, Università degli Studi di Urbino Carlo BO,
Urbino, Italy

Edutainment and Social 2026
16/05/2024 - 17/05/2024
Conference organiser, https://ludicomix.it/edutainment-social/
Empoli, Italy

Workshop: Perspectives on the Foundations of Information Theory and Statistical Mechanics
07/05/2026 - 09/05/2026
Organiser, Università della Svizzera Italiana,
Lugano, Switzerland

Edutainment and Social 2025
31/05/2025 - 02/06/2025
Conference organiser, https://ludicomix.it/edutainment-social/
Empoli, Italy

Workshop: Theoretical Equivalence and Related Topics
08/04/2025
Chair, Università Svizzera Italiana,
Lugano, Switzerland

VII Urbino Summer School in Epistemology
25/08/2025 - 28/08/2025
Organiser, Università degli Studi di Urbino Carlo BO,
Urbino, Italy

XXVIII International Summer School in Philosophy of Physics - Epistemology of Quantum Gravitiy
26/05/2025 - 30/05/2025
Organiser, Università degli Studi di Urbino Carlo BO,
Urbino, Italy

MOOC - Tempo, Universo e Particelle. Una introduzione alla filosofia della fisica
10/2023 - 12/2024
Coordinator, Università degli Studi di Urbino Carlo BO,
Urbino, Italy

VI Urbino Summer School in Epistemology
27/08/2024 - 30/08/2024
Organiser, Università degli Studi di Urbino Carlo BO,
Urbino, Italy

XXVII International Summer School in Philosophy of Physics - Philosophy of Dark Energy
03/06/2024 - 07/06/2024
Member of the Organising Committee, Università degli Studi di Urbino Carlo BO, Urbino, Italy

Edutainment and Social 2024
11/05/2024 - 12/05/2024
Conference organiser, https://ludicomix.it/edutainment-social/
Empoli, Italy

Reading Group Synergia
27/11/2023 - 31/03/2024
Reading Group organiser, Università degli Studi di Urbino Carlo BO, Urbino, Italy

Edutainment and Social 2023
28/05/2023
Conference organiser, https://www.ludicomix.it/edutainment-social/
Empoli, Italy

Membership

from 20/01/2025 Member of the **European Philosophy of Science Association (EPSA)**

from 05/09/2024 Member of the **The Italian Society for Logic and the Philosophy of Science (SILFS)**

from 01/11/2023 Member of the **Synergia Research Group**

Editorial Board

Editor Scientific Committee: **Sophia: Research on Fundamentals and Correlation of Knowledge**

Refereeing

Journals: **Foundations of Science**, **Foundations of Physics**, **Synthese**, **Isonomia**, **The Reasoner**, **Humana.mente**, **The European Physical Journal H**, **The British Journal for the Philosophy of Science**, **Humanities and Social Sciences Communications**, **Studies in History and Philosophy of Science**, **Philosophical Inquiries**